

Computer Architecture

What is a Computer?



It is an electronic device, composed of any digital circuits

Stores and processes data according to instructions given to it by a program stored in memory

Alan Turing

Father of Computer Science, Mathematician, Logician,
Wartime Codebreaker, Victim of Prejudice

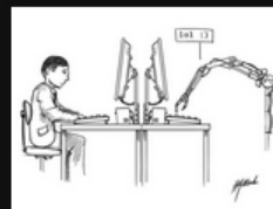


Criptography



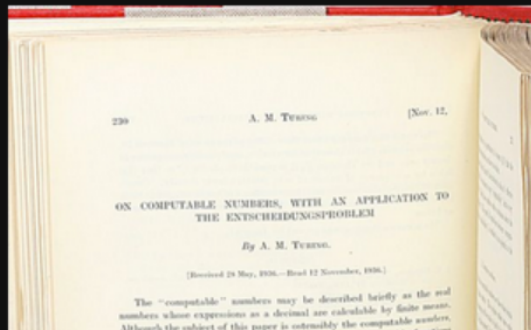
Turing's Bombe an electro-mechanical special purpose computing device used in the World War II effort

Artificial Intelligence



The Turing test

The Universal Computing Machine



The Turing Machine is an imaginary computer invented in this paper (1936) to help solve a question in mathematical logic.

The idea of controlling the machine's operations by means of a program of coded instructions stored in memory was born.

Electromechanical Computers

In the earliest computers, switches were opened and closed by magnetic fields produced by relays



The IBM ASCC was a general purpose electromechanical computer used in the World War II effort and built by a team representing Harvard, the U.S. Navy and IBM

Instructions and Data were provided via two separate punched paper tapes

One of the first programs to run on the Mark I was initiated on 29 March 1944 by [John von Neumann](#)

Modern Computers

How does the modern computer work?

The Von Neumann architecture

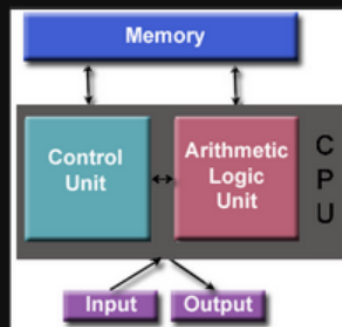


All computers based on the same basic design

Flexibility to execute different types of programs like the Turing machine

Program and Data both stored in the same memory

The Von Neumann architecture



- Stored-Program computer system
- Memory holds both the program data and the instructions processing it
- Control Unit manages moving data and program instructions into and out of memory
- Control Unit executes program instructions sequentially
- ALU is responsible for calculation tasks on the data
- Input/Output makes program interactive
- All these units are linked to each other via a computer bus

What is a computer bus?

Computer Bus

A bus, in computing, is a set of physical connections (cables, printed circuits, etc.) which can be shared by multiple hardware components in order to communicate with one another



Each wire in a computer bus will carry a single bit of information

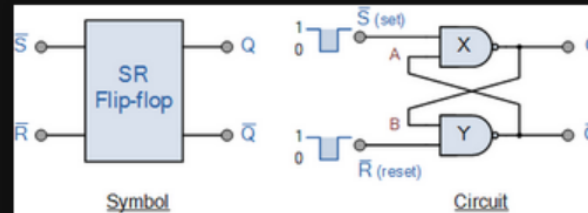
Memory Subsystem



Flip-Flops



In digital electronics, a flip-flop or latch is a circuit that has two stable states and can be used to store one bit of information

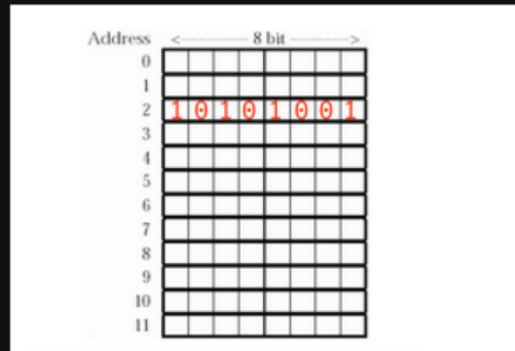


{ Board Exercise }

Draw the truth table for an SR Flip-Flop

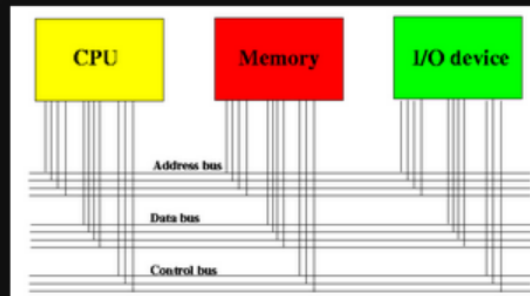
Memory Organisation

Main Memory is organised as a matrix of bits, where each row represents a memory location and is capable of storing a number of a specific width, typically multi-byte in length (8, 16, 32 or 64 bits)



Assume that data within memory can only be read or written a single row (memory location) at a time

Memory Architecture



- Address Bus - carries the address of a memory cell
- Data Bus - used to transfer data to and from a memory cell
- Control Bus - selects read or write operations between memory and cpu or I/O devices

Random Access Memory



- Many memory cells of fixed size
- Each cell has an address associated with it
- The time it takes to fetch/store is the same for all cells
- Contents are lost if power fails

Read Only Memory



- Contents written during production
- Supports read operations only
- Contents are NOT lost if power fails
- Later evolved into PROMs, EPROMs and EEPROMs

Memory stores numbers, but what about letters?

Data in our programs can consist of both Numbers and Characters

But how can a computer represent Characters?

Using a character encoding scheme

ASCII

American Standard Code for Information Interchange



- Most common character-encoding scheme on the Web until 2008
- Surpassed by UTF-8, which includes ASCII as a subset
- Based on the English alphabet, uses 7 bits and is capable of encoding:
 1. numbers 0 to 9
 2. lowercase letters a to z
 3. uppercase letters A to Z
 4. basic punctuation symbols
 5. control codes that were originated with Teletype machines

{ Exercise }

Consult ASCII Man Page

The Memory Subsystem

Storing Data in Memory

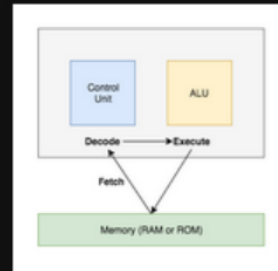
What about more complex things?

Anything can be represented in memory using numbers, we just need some structure and/or encoding mechanism

.gif | .mp3 | .wav | .avi | .psd | .jpeg

The Central Processing Unit

- Executes a program of instruction cycles
- Each instruction cycle has a number of steps



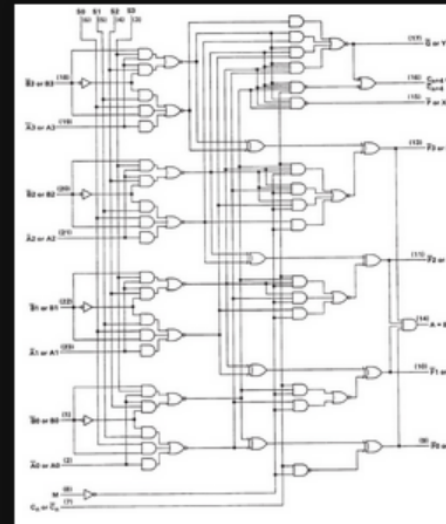
1. Fetch an instruction from memory
2. Decode it, determine what has to be done
3. Execute the instruction
4. Go back to step 1

ALU

SELECTION				ACTIVE-HIGH DATA		
				M = H LOGIC FUNCTIONS	M = L: ARITHMETIC OPERATIONS	
S3	S2	S1	S0		$C_n = H$ (no carry)	$C_n = L$ (with carry)
L	L	L	L	$F = A$	$F = A$	$F = A \text{ PLUS } 1$
L	L	L	H	$F = A \oplus B$	$F = A \oplus B$	$F = (A \oplus B) \text{ PLUS } 1$
L	L	H	L	$F = \overline{A}B$	$F = A \oplus \overline{B}$	$F = (A \oplus \overline{B}) \text{ PLUS } 1$
L	L	H	H	$F = 0$	$F = \text{MINUS } 1 \text{ (2's COMPL.)}$	$F = \text{ZERO}$
L	H	L	L	$F = \overline{A\overline{B}}$	$F = A \text{ PLUS } A\overline{B}$	$F = A \text{ PLUS } A\overline{B} \text{ PLUS } 1$
L	H	L	H	$F = \overline{B}$	$F = (A \oplus B) \text{ PLUS } A\overline{B}$	$F = (A \oplus B) \text{ PLUS } A\overline{B} \text{ PLUS } 1$
L	H	H	L	$F = A \ominus B$	$F = A \text{ MINUS } B \text{ MINUS } 1$	$F = A \text{ MINUS } B$
L	H	H	H	$F = A\overline{B}$	$F = A\overline{B} \text{ MINUS } 1$	$F = A\overline{B}$
H	L	L	L	$F = \overline{A} \oplus B$	$F = A \text{ PLUS } AB$	$F = A \text{ PLUS } AB \text{ PLUS } 1$
H	L	L	H	$F = A \oplus \overline{B}$	$F = A \text{ PLUS } B$	$F = A \text{ PLUS } B \text{ PLUS } 1$
H	L	H	L	$F = B$	$F = (A \oplus \overline{B}) \text{ PLUS } AB$	$F = (A \oplus \overline{B}) \text{ PLUS } AB \text{ PLUS } 1$
H	L	H	H	$F = AB$	$F = AB \text{ MINUS } 1$	$F = AB$
H	H	L	L	$F = 1$	$F = A \text{ PLUS } A^1$	$F = A \text{ PLUS } A \text{ PLUS } 1$
H	H	L	H	$F = A \oplus \overline{B}$	$F = (A \oplus B) \text{ PLUS } A$	$F = (A \oplus B) \text{ PLUS } A \text{ PLUS } 1$
H	H	H	L	$F = A \oplus B$	$F = (A \oplus \overline{B}) \text{ PLUS } A$	$F = (A \oplus \overline{B}) \text{ PLUS } A \text{ PLUS } 1$
H	H	H	H	$F = A$	$F = A \text{ MINUS } 1$	$F = A$

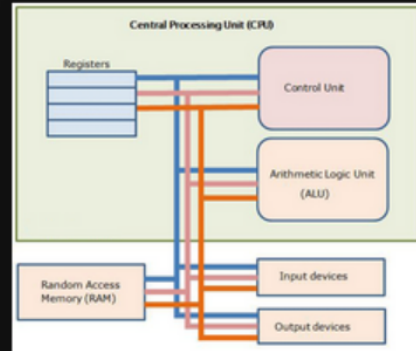
¹ Each bit is shifted to the next more significant position.

4 bit Cascadable ALU with 16 different Operations, implemented using 75 Logic Gates.



CPU Registers

Used to store and transfer data and instructions, they are the fastest type of memory available.



Some registers are general purpose, others have specific functions. The accumulator register, for example, is located inside the ALU and used during arithmetical & logical operations

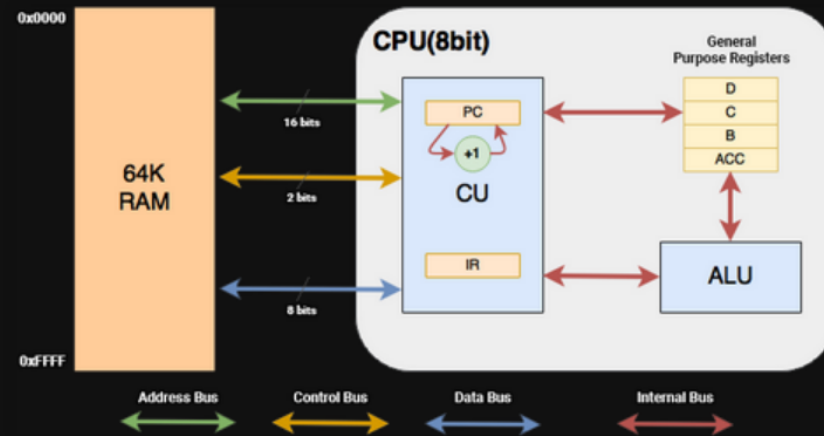
Machine Code and Assembly

Machine code or machine language is a set of instructions executed directly by a computer's CPU. Each instruction performs a very specific task, such as a load, a jump, or an ALU operation on a unit of data in a CPU register or memory.

Machine code	Assembly code	Description
001 1 000010	LOAD #2	Load the value 2 into the Accumulator
010 0 001101	STORE 13	Store the value of the Accumulator in memory location 13
001 1 000101	LOAD #5	Load the value 5 into the Accumulator
010 0 001110	STORE 14	Store the value of the Accumulator in memory location 14
001 0 001101	LOAD 13	Load the value of memory location 13 into the Accumulator
011 0 001110	ADD 14	Add the value of memory location 14 to the Accumulator
010 0 001111	STORE 15	Store the value of the Accumulator in memory location 15
111 0 000000	HALT	Stop execution

Because strings of 0s and 1s are so hard for people to read, assembly languages were invented. An assembly language is essentially a human readable direct translation of machine language.

Program Execution



- Data flows inside the CPU using an internal bus
- IR holds the next instruction to be decoded and executed
- PC holds the address of the next instruction to be fetched

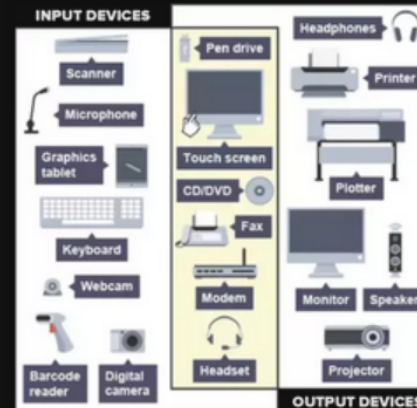
{ Board Exercise }

Simulate the machine code execution in the Von Neumann architecture

I/O

There are different types of I/O

- Input Devices
 - Keyboard
 - Mouse
 - Microphone
- Output Devices
 - Monitor
 - Printer
 - Speakers
- Secondary Memory
 - Hard Drive
 - Floppy Disk Drive
 - CD/DVD



Terminal Evolution



teletype



old terminal



modern terminal

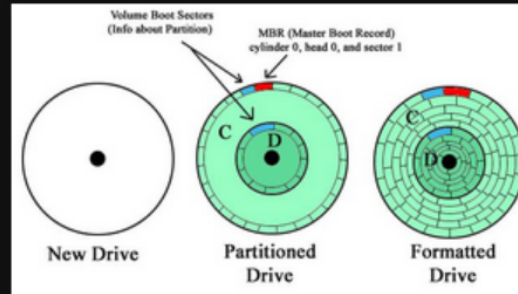
Storage Memory

Also called Secondary memory, can be Removable or Persistent



- Much slower than RAM/ROM
- Much bigger than RAM/ROM
- Much cheaper than RAM/ROM
- Used by the OS through filesystems
- Can be used by the OS as virtual memory

Hard Disk Structure



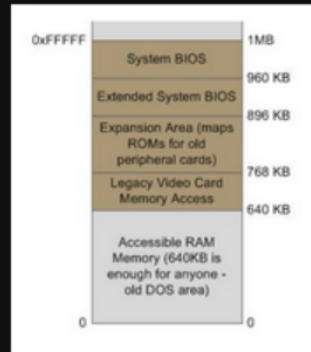
- MBR is used in the computer boot process
- Drives can be split into logical volumes called partitions
- Sectors for a disk partition are marked during low level formatting
- An empty file system is installed on a disk partition during the high level formatting

Computer Boot Sequence

Power On

The PC is set to an address 16 bytes below the processor's uppermost physical address. Eg: 0xFFFF0 for a 1M of address space

ROM containing the software initialization code must be located at this address

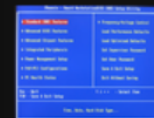


Basic Input/Output System

BIOS is a Boot Firmware built into PCs and the first software to be run at power on

Performs hardware initialization (POST) and loads a Bootloader from a mass memory device

Unified Extensible Firmware Interface (UEFI) was designed as a successor to BIOS, aiming to address its technical shortcomings. New PC hardware predominantly ships with UEFI firmware



The Boot Process

